Here are some notes on estimation theory
I use R, a free statistical software package. See http://www.r-project.org/
Let's generate 10 observations from an exponential distribution with parameter 2.

> rexp(10,2)
[1] 0.032090372 0.846786086 0.037242202 0.868357479 0.091108944 0.001468721 [7] 1.141715637 0.368750720 0.172478341 0.027039513

Do it yet another time, then we get different values:

> rexp(10,2)
[1] 0.64568820 0.85705537 0.38006965 1.12513635 0.71712269 0.70836800 [7] 0.33664569 0.34383845 0.31659871 0.03829912

Each time we see different values – the first sample differs from the second one. This is because the observations are randomly generated, independently of each other.
They are realizations of random variables!
Let's generate once more 10 observations from this distribution and store it as a vector x:

> x<-rexp(10,2) >x

[1] 0.65758437 1.22034032 0.06914319 0.98559390 0.28810438 0.28761869 [7] 0.65574415 0.18215762 0.23036037 0.74237939

We can compute its mean

> mean(x)
[1] 0.5319026

Compute the mean of a sample of size 100:

> mean(rexp(100,2)) [1] 0.632066

of size 1000
> mean(rexp(1000,2)) [1] 0.4840925

of size 10000:
> mean(rexp(10000,2)) [1] 0.5010878

and size 100000
> mean(rexp(100000,2)) [1] 0.4999304

What do we see: the averages converge to the true mean of an exponential distribution of parameter 2. Remember that E(X)=1/2 indeed.

This is a consequence of the law of large numbers.
Consequently, we can take as an estimate for the parameter 1/(sample average) and with high probability, this will be close to the true parameter value.

Each time we generate a new sample we get different outcomes for the observations, means and thus for the estimates.

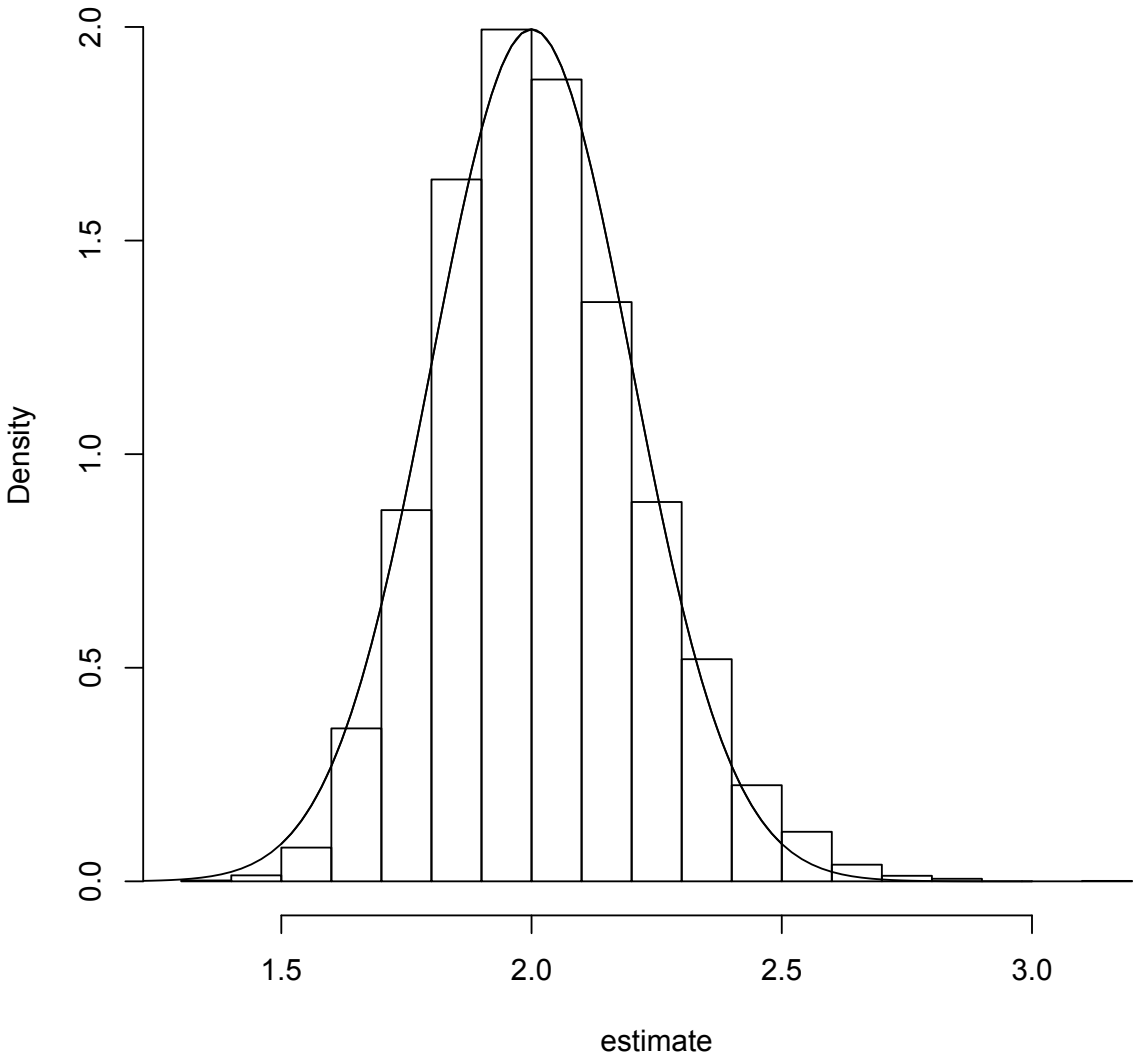
Now we assess the variability of the estimates. Let's try this via simulation: We generate 10000 samples. For each sample we compute the sample mean and the estimate 1/(sample mean):

> estimate<-c(1:10000)
> for (i in 1:10000) { estimate[i]<-1/mean(rexp(100,2)) }

and a histogram of the 10000 values of the estimates:
> h<-hist(estimate,plot=F)
We superimpose the density of a normal with mean 2 and standard deviation 2/10:

> hist(estimate,freq=F)
> curve(dnorm(x,2,.2), from=1.5, to= 3.0, add=TRUE)

We will see soon that the mean of our estimator is close to 2, and its variance close to 4/100. In fact, it looks as if our estimator has approximately an normal distribution with mean 2 and variance 4/100. The role of the normal distribution will be explained later as well.


Histogram of estimate
Density
0.0
0.5
1.0
1.5
2.0
1.5
2.0
2.5
3.0
estimate